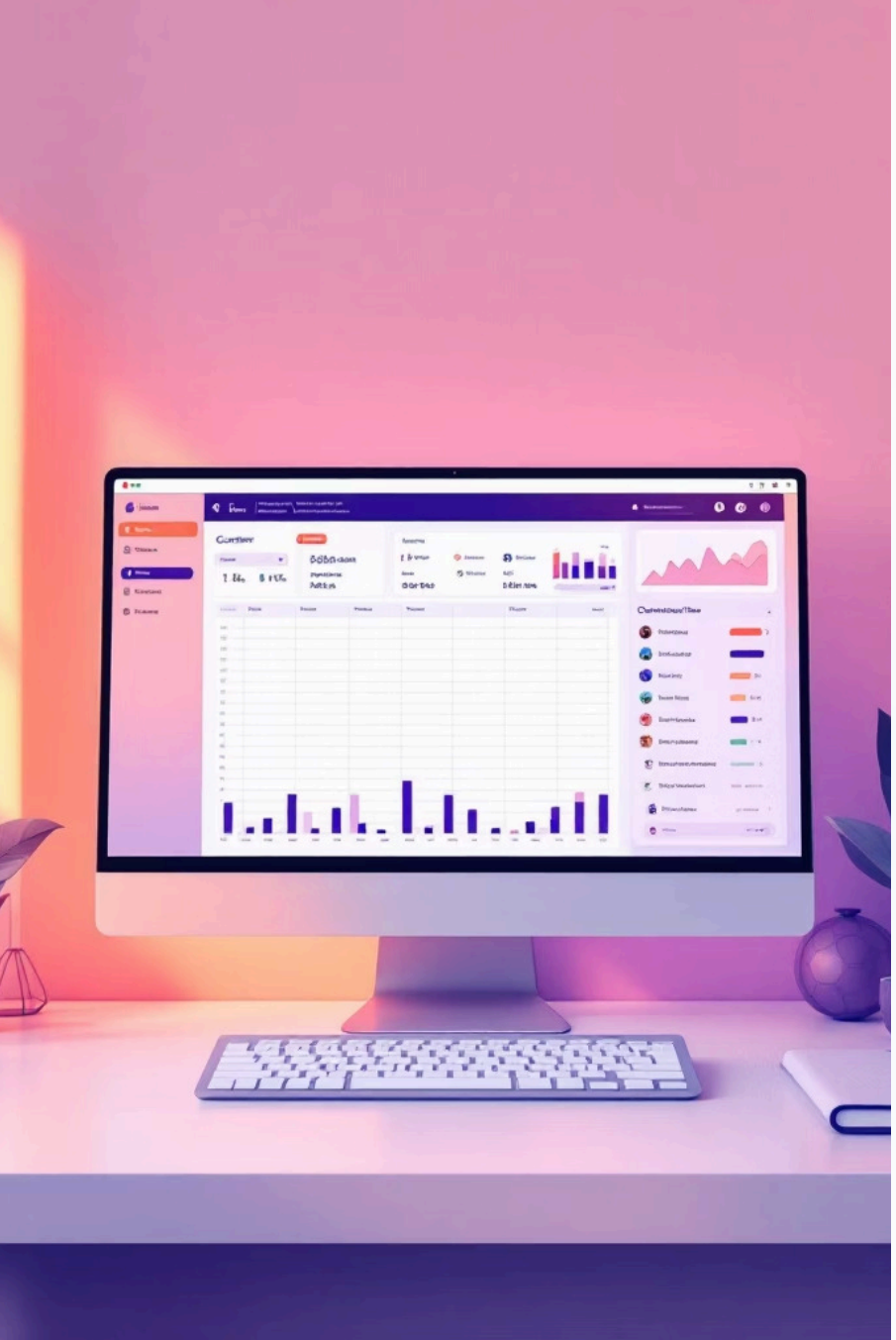
Customer Shopping Behavior Analysis

Insights from 3,900 transactions to drive strategic business decisions

Made with GAMMA



Dataset Overview

3,900

Purchase Records

Complete transactional data analyzed

18

Data Columns

Comprehensive customer and purchase details

37

Missing Values

Missing values in Review Rating column

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Data Preparation Process

01

Data Loading

Imported dataset using pandas library

02

Initial Exploration

df.info() and df.describe() for structure and statistics

03

Missing Values

Median rating by category for Review Rating column

04

Standardization

Converted columns to snake_case format

05

Feature Engineering

Created age_group and purchase_frequency_days columns

06

Database Integration

Loaded into Microsoft SQL Server for analysis

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SQL Analysis: Revenue Insights

Revenue by Gender

Compared total revenue from male and female customers

Shipping Comparison

Analyzed Standard vs. Express shipping purchase values

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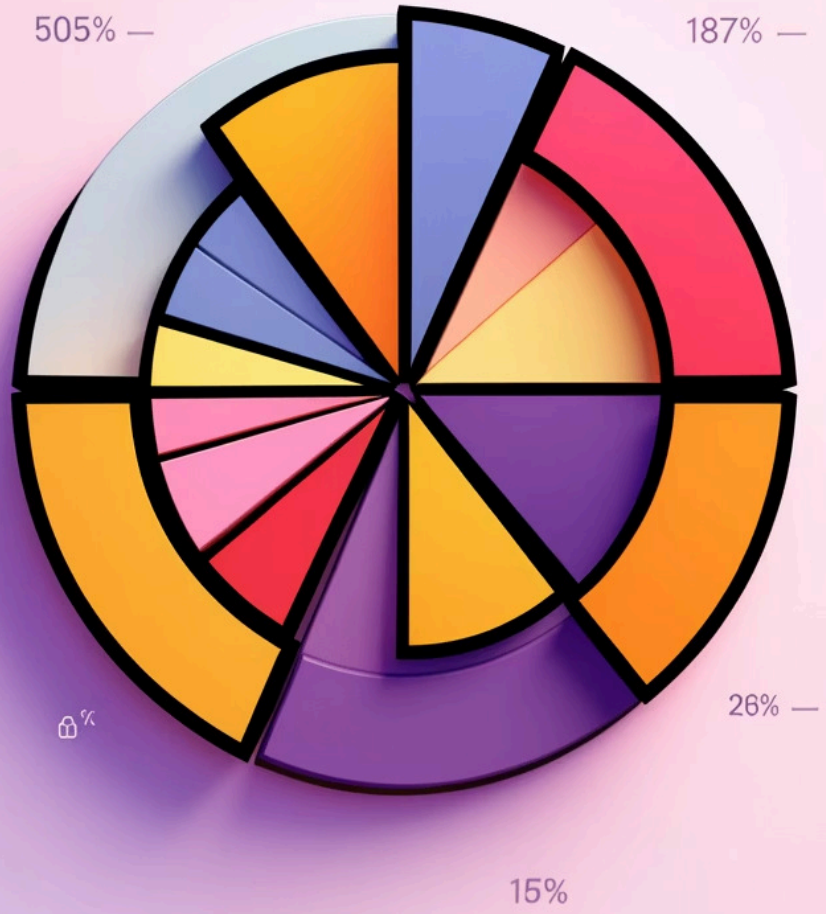

SQL Analysis: Customer Segmentation

Top Products by Rating

Products with highest average customer ratings

Customer Segments

New, Returning, and Loyal customer categories

Discount-Dependent Products

Products with highest proportion of discounted purchases

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SQL Analysis: Subscription Analysis

Subscribers vs. Non-Subscribers

Compared average spending and total revenue

Repeat Buyers

Customers with >5 purchases and subscription likelihood

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SQL Analysis: Product Performance

1

Top 3 Products per Category

Most frequently purchased products in each category

2

High-Spending Discount Users

Customers who applied discounts but spent above average

3

Revenue by Age Group

Total revenue generated by different age groups

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Power BI Dashboard

Interactive visualizations for trends, comparisons, and key performance indicators

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Business Recommendations

Boost Subscriptions

Exclusive benefits and incentives

Customer Loyalty Programs

Reward systems for repeat customers

Review Discount Policy

Balance sales and profit margins

Product Positioning

Promote top-rated products effectively

Targeted Marketing

Focus on high-revenue age groups and express shipping customers

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Key Takeaways

Comprehensive Analysis

3,900 transactions analyzed across 18 data points

Strategic Insights

Revenue patterns, customer segments, and product performance identified

Actionable Recommendations

Five strategic initiatives to drive business growth

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